

SCOPE OF WORK (SOW)

**Recreation Center Renovation Phase III - HVAC (Air Conditioning &
Mechanical Ventilation Systems)**

1. GENERAL REQUIREMENTS

1.1 Scope

1. Vendor shall strictly follow all details indicated in the contract drawings and all field conditions observed during the mandatory site visit and shall reconcile both to deliver a fully functional air conditioning system serving a clubhouse with an average occupancy load of 120 persons.
2. Vendor shall be solely responsible for verifying that the proposed system design, capacities, and configurations meet the project requirements and performance criteria in accordance with the specified standards.
3. Vendor shall review all design documents and specifications in detail and raise any discrepancies, omissions, or technical concerns prior to submission of the quotation.
4. In the event of award, the Vendor shall again review and confirm all design aspects prior to commencement of work and coordinate with the Contracting Officer/Engineer for resolution of any identified issues.
5. Vendor shall bear full responsibility for achieving a complete, fully operational, and compliant HVAC system and shall make all necessary adjustments, modifications, and refinements to the design and installation to meet performance requirements.
6. Vendor shall execute all ancillary, incidental, and minor works necessary for proper system functionality, whether or not explicitly indicated, at no additional cost where such works fall within five percent (5%) of the total contract value.
7. Vendor's obligation includes ensuring system performance, reliability, and integration, and no claim for additional compensation shall be entertained for works required to achieve a fully functional system within the defined scope.

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1.2 Provide all labor, supervision, materials, tools, equipment, and incidentals required for a **complete, operable HVAC system**, including:

- VRV/VRF air conditioning system
- Mechanical ventilation system
- Controls, power wiring coordination
- Testing, Adjusting & Balancing (TAB)
- Commissioning

Work shall be turnkey and include all items whether specifically mentioned or required for proper system operation.

2. APPLICABLE CODES AND STANDARDS

All work shall comply with latest editions of:

- **ASHRAE Standards (62.1, 90.1)**
 - **SMACNA Duct Construction Standards**
 - **International Code Council – International Mechanical Code (IMC)**
 - **National Fire Protection Association – NFPA 70 (NEC)**
 - **Air Conditioning Contractors of America Manual D (duct design)**
 - UL listed equipment and components
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3. SUBMITTALS

Contractor shall submit for approval prior to installation:

- Equipment data sheets (capacity, power, performance curves)
- Shop drawings (ducting, piping, layouts, clearances)
- Refrigerant pipe sizing calculations
- Static pressure and airflow calculations

- Control schematics
 - Product certificates and compliance documentation
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4. VRV/VRF AIR CONDITIONING SYSTEM

4.1 System Design Requirements

- System shall be **inverter-driven VRV/VRF type**
 - Capable of simultaneous multi-zone conditioning
 - Minimum COP/EER as per ASHRAE 90.1
 - Refrigerant: R410A or equivalent approved
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4.2 Indoor Units

Install:

- 48,000 BTU/hr ceiling cassette units (4 Nos.)
- 24,000 BTU/hr floor-mounted units (3 Nos.)
- 24,000 BTU/hr concealed duct unit (1 No.)

Technical Requirements:

- Sound level ≤ 45 dB(A) (indoor occupied zones)
 - Built-in drain pumps (where applicable)
 - Washable air filters (MERV 8 minimum)
 - Wireless or wall-mounted controllers
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4.3 Outdoor Units

Install VRV condensing units (2 Nos.):

- Microprocessor-controlled inverter compressors

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- Ambient operating range: 20°F to 122°F (or better)
 - Anti-corrosion coated coils
 - Low-noise operation ≤ 65 dB(A)
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4.4 Refrigerant Piping System

Materials:

- Type L hard-drawn copper (ASTM B280 compliant)

Installation Requirements:

- Brazed joints using nitrogen purging
- Proper oil return slopes
- Isolation supports at 6 ft spacing (max)
- Identification labels at all branches

Testing:

- Pressure test: 500 psi (dry nitrogen) for 24 hrs.
 - Vacuum: ≤ 500 microns prior to charging
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4.5 Insulation System

- Refrigerant piping:
 - Inner: Closed-cell nitrile rubber
 - Outer: Glass wool wrap
 - External: UV-resistant polyethylene cladding
 - Minimum thickness: 1 inch
 - Vapor barrier required on all chilled lines
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4.6 Condensate Drain System

- PVC Schedule 40 piping
 - Minimum slope: 1% (1/8" per ft)
 - Include:
 - Cleanouts
 - Trap seals
 - Air gap at discharge
 - Insulation:
 - 1-inch nitrile rubber (anti-condensation)
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4.7 Fresh Air Integration

- Provide treated or untreated fresh air supply as indicated
 - Minimum ventilation rates per ASHRAE 62.1
 - Include:
 - Volume control dampers
 - Backdraft dampers
 - Air balancing provisions
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5. MECHANICAL VENTILATION SYSTEM

5.1 Exhaust Fans

- Inline centrifugal fans (2 Nos.)
- Capacity: 50 CFM @ 100 Pa
- Power: 120V/1Ph/60Hz
- Features:

- Thermal overload protection
 - Backdraft damper
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5.2 Ductwork System

Materials:

- Galvanized steel ducts (0.6 mm thickness)

Construction:

- As per SMACNA low-pressure standards
- Transverse joints sealed (Class A sealing)
- Leakage Class ≤ 6

Supports:

- Hangers at ≤ 8 ft intervals
 - Vibration isolation where required
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5.3 Flexible Ducts

- Aluminum flexible ducts (110 mm Dia)
 - Max length: 3 m
 - Fully stretched installation required
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5.4 Air Devices

- Powder-coated aluminum louvers with bird screens
 - Disc valves (adjustable airflow type)
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6. ELECTRICAL & CONTROLS

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- Coordinate with electrical contractor for power supply
 - Provide:
 - Control wiring
 - Interlocks with lighting (toilet exhaust fans)
 - All wiring shall comply with:
 - NEC requirements
 - Conduit protection
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7. TESTING, ADJUSTING & BALANCING (TAB)

Perform TAB in accordance with **Associated Air Balance Council standards**:

- Airflow balancing within $\pm 10\%$ of design
- Verify:
 - Supply air volumes
 - Exhaust air volumes
 - Static pressures

Provide certified TAB report.

8. COMMISSIONING

- Functional performance testing of all systems
- Demonstrate:
 - Cooling capacity
 - Controls operation
 - Fail-safe modes
- Provide training to Owner personnel

9. QUALITY ASSURANCE

- Minimum 5 years experience for HVAC contractor
 - Installers shall be certified technicians
 - All equipment shall be new and unused
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10. WARRANTY

- Minimum:
 - 1 year workmanship warranty
 - 5 years compressor warranty
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11. CLOSEOUT REQUIREMENTS

Provide:

- As-built drawings
 - O&M manuals
 - Warranty certificates
 - Spare parts list
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12. ACCEPTANCE CRITERIA

System shall be accepted only upon:

- Successful commissioning
- Compliance with airflow and cooling performance
- Zero refrigerant leakage
- No abnormal vibration/noise

13. Warranty for the System and Workmanship

1. Warranty for any individual item or system shall be for two (2) years or the manufacturer's standard warranty period, whichever is greater.
2. Vendor shall provide a written functional warranty for the complete HVAC system for a period of two (2) years from the date of final acceptance.
3. During the warranty period, the Vendor shall repair or replace, at no cost to the Embassy, any defective equipment, components, or workmanship identified.
4. The warranty shall cover all parts, labor, and associated services required to restore the system to full operational condition.
5. Vendor shall respond to warranty service requests within a reasonable timeframe and ensure prompt rectification of any issues to maintain continuous system functionality